

10.6 Solution: (a) For incident wave with polarization

$$\frac{\hat{\mathbf{e}}_x \pm i\hat{\mathbf{e}}_y}{\sqrt{2}},$$

the scattered electric field is, by Eq. (10.57),

$$\mathbf{E}_{sc} = \sqrt{\frac{\pi}{2}} \sum_{\ell=1}^{\infty} i^{\ell} \sqrt{2\ell+1} \left[\alpha_{\ell}(\omega) h_{\ell}^{(1)}(kr) \mathbf{X}_{\ell,\pm 1} \pm \frac{\beta_{\ell}(\omega)}{k} \nabla \times \left(h_{\ell}^{(1)}(kr) \mathbf{X}_{\ell,\pm 1} \right) \right].$$

Notice the difference of a $1/\sqrt{2}$ factor. Using Eq. (10.60), the second term becomes

$$\nabla \times \left(h_{\ell}^{(1)}(kr) \mathbf{X}_{\ell,\pm 1} \right) = i\hat{\mathbf{n}} \frac{\sqrt{\ell(\ell+1)}}{r} h_{\ell}^{(1)}(kr) Y_{\ell,\pm 1} + \frac{i}{r} \frac{\partial}{\partial r} \left[r h_{\ell}^{(1)}(kr) \right] \hat{\mathbf{n}} \times \mathbf{X}_{\ell,\pm 1}.$$

Now, using the asymptotic expression for

$$h_{\ell}^{(1)}(kr) = (-i)^{\ell+1} \frac{e^{ikr}}{kr},$$

the scattered field is

$$\mathbf{E}_{sc} = \frac{e^{ikr}}{r} \sqrt{\frac{\pi}{2}} \frac{1}{k} \sum_{\ell=1}^{\infty} \sqrt{2\ell+1} \left[\alpha_{\ell}(\omega) \mathbf{X}_{\ell,\pm 1} \pm i\beta_{\ell}(\omega) \frac{\sqrt{\ell(\ell+1)}}{r} Y_{\ell,\pm 1} \hat{\mathbf{n}} - i\beta_{\ell}(\omega) \hat{\mathbf{n}} \times \mathbf{X}_{\ell,\pm 1} \right].$$

By the definition of $\mathbf{f}(\hat{\mathbf{k}}, \hat{\mathbf{k}}_0)$ and also notice that $E_0 = 1$, we obtain

$$\mathbf{f} = \frac{1}{ik} \sqrt{\frac{\pi}{2}} \sum_{\ell=1}^{\infty} \sqrt{2\ell+1} [\alpha_{\ell}(\omega) \mathbf{X}_{\ell,\pm 1} \pm i\beta_{\ell}(\omega) \hat{\mathbf{n}} \times \mathbf{X}_{\ell,\pm 1}].$$

(b) By the optical theorem,

$$\sigma_t = \frac{4\pi}{k} \text{Im} \left[\hat{\mathbf{e}}_0^* \cdot \mathbf{f}(\hat{\mathbf{k}} = \hat{\mathbf{k}}_0) \right].$$

Here,

$$\hat{\mathbf{e}}_0 = \frac{\hat{\mathbf{e}}_x \pm i\hat{\mathbf{e}}_y}{\sqrt{2}}.$$

From solution to Prob. 10.2, we have

$$\mathbf{X}_{\ell m} = \frac{1}{\sqrt{\ell(\ell+1)}} \begin{pmatrix} \frac{1}{2} \left[\sqrt{(\ell-m)(\ell+m+1)} Y_{\ell,m+1} + \sqrt{(\ell+m)(\ell-m+1)} Y_{\ell,m-1} \right] \\ \frac{1}{2i} \left[\sqrt{(\ell-m)(\ell+m+1)} Y_{\ell,m+1} - \sqrt{(\ell+m)(\ell-m+1)} Y_{\ell,m-1} \right] \\ m Y_{\ell m} \end{pmatrix}.$$

For $m = +1$,

$$\mathbf{X}_{\ell,+1} = \frac{1}{\sqrt{\ell(\ell+1)}} \begin{pmatrix} \frac{1}{2} \left[\sqrt{(\ell-1)(\ell+2)} Y_{\ell,2} + \sqrt{\ell(\ell+1)} Y_{\ell,0} \right] \\ \frac{1}{2i} \left[\sqrt{(\ell-1)(\ell+2)} Y_{\ell,2} - \sqrt{\ell(\ell+1)} Y_{\ell,0} \right] \\ Y_{\ell,1} \end{pmatrix}.$$

For $m = -1$,

$$\mathbf{X}_{\ell,-1} = \frac{1}{\sqrt{\ell(\ell+1)}} \begin{pmatrix} \frac{1}{2} \left[\sqrt{\ell(\ell+1)} Y_{\ell,0} + \sqrt{(\ell-1)(\ell+2)} Y_{\ell,-2} \right] \\ \frac{1}{2i} \left[\sqrt{\ell(\ell+1)} Y_{\ell,0} - \sqrt{(\ell-1)(\ell+2)} Y_{\ell,-2} \right] \\ - Y_{\ell,-1} \end{pmatrix}.$$

For $\hat{\mathbf{k}} = \hat{\mathbf{k}}_0$, $\theta = \phi = 0$, using the special value of the spherical harmonics,

$$Y_{\ell m}(0, \phi) = \sqrt{\frac{2\ell+1}{4\pi}} \delta_{m,0},$$

we have

$$\begin{aligned} \mathbf{X}_{\ell,+1}(0,0) &= \frac{\sqrt{2\ell+1}}{4\sqrt{\pi}} (1, i, 0)^\top, \\ \mathbf{X}_{\ell,-1}(0,0) &= \frac{\sqrt{2\ell+1}}{4\sqrt{\pi}} (1, -i, 0)^\top. \end{aligned}$$

Also,

$$\hat{\mathbf{n}} = \hat{\mathbf{e}}_z = (0, 0, 1),$$

then

$$\begin{aligned} \hat{\mathbf{n}} \times \mathbf{X}_{\ell,+1}(0,0) &= \frac{\sqrt{2\ell+1}}{4\sqrt{\pi}} (-i, 1, 0)^\top, \\ \hat{\mathbf{n}} \times \mathbf{X}_{\ell,-1}(0,0) &= \frac{\sqrt{2\ell+1}}{4\sqrt{\pi}} (i, 1, 0)^\top. \\ \hat{\mathbf{e}}_+^* \cdot \mathbf{X}_{\ell,+1} &= \frac{\sqrt{2\ell+1}}{4} \sqrt{\frac{2}{\pi}}, \\ \hat{\mathbf{e}}_+^* \cdot (\hat{\mathbf{n}} \times \mathbf{X}_{\ell,+1}) &= -i \frac{\sqrt{2\ell+1}}{4} \sqrt{\frac{2}{\pi}}, \\ \hat{\mathbf{e}}_-^* \cdot \mathbf{X}_{\ell,-1} &= \frac{\sqrt{2\ell+1}}{4} \sqrt{\frac{2}{\pi}}, \\ \hat{\mathbf{e}}_-^* \cdot (\hat{\mathbf{n}} \times \mathbf{X}_{\ell,-1}) &= i \frac{\sqrt{2\ell+1}}{4} \sqrt{\frac{2}{\pi}}. \end{aligned}$$

Therefore,

$$\begin{aligned} \sigma_t &= \frac{4\pi}{k} \frac{1}{k} \sqrt{\frac{\pi}{2}} \sum_{\ell=1}^{\infty} \text{Im} \left[\frac{1}{4i} (\alpha_\ell(\omega) + \beta_\ell(\omega)) \right] (2\ell+1) \sqrt{\frac{2}{\pi}} \\ &= -\frac{\pi}{k^2} \sum_{\ell=1}^{\infty} (2\ell+1) \text{Re} [\alpha_\ell(\omega) + \beta_\ell(\omega)]. \end{aligned}$$